

Chris Frith ~ Active Inference Insights 009 ~ Mentalizing, Schizophrenia, C

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hello everyone and welcome back to
active inference insights I your host
darus parvey Wayne and today I
absolutely thrilled to be speaking with
Professor Chris FFF Chris is a
neuropsychologist and a Pioneer in the
application of brain Imaging to the
study of mental processes he has
contributed more than 500 papers to
peer-reviewed Scientific journals and is
known especially for his work on agency
schizophrenia and social cognition
within this latter he has worked extens
extensively with his wife Professor
utaff together they have published
seminal papers on theory of mind and
social signaling and last year they
published the book what makes us social
again it is no overstatement to say that
Chris possesses a legendary status in
the Spheres of Psychology and
Neuroscience and so it is a real Honor
on my part to be chatting to him today
Chris thank you so much for joining me
and welcome to the show oh thanks for
those nice words you're welcome uh it's
are you are you in London currently it's
a beautiful day
yes all be a bit Frosty yeah uh
excellent so as I mentioned in my
introduction uh you root many or you
ground many of our social capacities in
mentalizing which requires theory of
mind now some of our audience might not
we we our audience is diverse you know

act brings in people from maths physics
all sorts they might not have a
psychology background so they might not
be aware of what mentalizing or theory
of Mind actually is so it might be worth
just setting the stage and giving a
brief exposition of what those terms
mean if possible right oh yes theory of
mind is a rather
misleading

phrase because it's not a theory of how
the mind works and it's probably not
even a theory so mentalizing is a
slightly better term I think but the
idea

is
that a one way into this is to say we're
trying to predict prct is useful to be
able to predict what people are going to
do and to understand why they're doing
it and theory of Mind originally said
people do things because of their hidden
mental state such as beliefs and
intentions so if somebody takes an
umbrella with them this doesn't mean
that it's raining it means that they
believe it is reigning or is going to
rain and the

key um
to the experiments on people's ability
to understand these hidden mental States
is the impact of false
beliefs because if somebody if you know
it's not raining and this person goes
out with an umbrella then you can say he
has a false belief that it's going to
rain and his belief is different from
mine and I have to take this into

account when I predict what he is going to do so that's basically what mentalizing is about that's the process of predicting what other people's mental states are and mental States meaning things like intentions beliefs and desires and we think that this is probably different from understanding people's emotions yes I can imagine a lot of people make that conflation um I guess and a further thing to unpack there is when people hear about sort of mentalizing they might think of autism and and the associated difficulties in mentalizing that seem to be associated with autism but over time uh many autistic people use heuristics or tricks or social cues to mentalize yes is there a difference because behaviorally they might manifest similarly and permit kind of harmonious social interaction or is theory of mind or mentalizing committed to saying that uh those who are mentalizing actually have a mental representation a kind of recursive mental representation of what the other part in this thinking um well that's what I think I don't think everybody agrees with that there is I mean more recently there is evidence that there are two I mean depends on how you decide to talk about it two levels of mentalizing which some people say the low one is not really mentalizing but that is to say some sometimes called Behavior

reading um that you can solve many you
can make many of these predictions
without needing to know about other
people's mental

States and um this is and it's so all
also in some sense automatic so it
happens you don't have to think about it
you

just um know this is the right way to
behave next in in this interaction and
in relation to autism there's some
suggestion that it's this lower level
part that they find difficult and that
the her istics as you call them are in
fact something that we all use so
there's this there's this thing
called folk

psychology which is supposed to be
inferior but I think is actually very
important which is that we know this is
the idea that people do indeed behave
because of their beliefs and desires and
that's how we can predict and their
behavior and this is a cultural
phenomena so in different cultures you
may have different ideas about how folk
psychology works and in particular in
um there's some anthropological work on
the C in people who come from the
Pacific Rim which includes some Meers in
Mexico and Fiji and these various other
places where they there's a
cultural attitude that it's impossible
to know what other people's mental
states

are and that you shouldn't bother to try
and in fact children are brought up not
you know you shouldn't be doing this

sort of thing you just look at their behavior and it even extends to sort of interesting semi-legal situations where in the west we're very concerned that you know it's the intention that counts not it's more important the intention is more important than the outcome whereas in these other cultures the outcome is all that matters and even if it's an accident you'll be punished interesting there's an interesting level of I think cultural effects on some of these mentalizing stories yes philosophically I certainly have issues with that but we won't we won't go into that too much um yes that's interesting I guess because it also maps onto our scientific institutions and our way of actually viewing what's going on mentally um EXA I can't help but notice that we are presupposing mental representations even when I asked that question and I've been sort of delving into the world of radical inactivist cognition and they would say well there are no representations there are there is no content all we have are these Dynamic relationships between or these Dynamic couplings uh between agent and Arena right does that is that does theory of Mind collapse if we just permit that these are um I guess they talk about sort of perceiving correlations in the environment that's something that they're very excited about but actually this isn't a in you know this isn't necessarily a representation or intentional stance

um does it does it is it compatible with
these sort of ramped up versions of an
activist

cognition hello everyone and I do
apologize for interrupting the flow of
the conversation with Chris I wanted to
add this little bit because I realized
uh upon watching the conversation that I
had with Chris that this juncture I did
misspeak and misconstrue and
overgeneralize the claims of wreck at
least explicated by hat and Mayan so I
made it out as if uh they say that all
cognition invol uh is contentless this
is not actually what they say they say
that basic mind or basic cognition is
contentless whereby basic cognition
means all cognitive activities except
those involving public Language and
Cultural symbol systems so again they do
not say that all cognition is
contentless because they because there
are also these higher order forms of
cognition which might be content
involving and again those require public
symbols social practices of symbolic uh
communication so they're therefore
governed by semantic norms and these
correctness conditions so where the
theory of mind is that uh that higher
order form or lower order form I guess
is for you to decide but you'll see
later on I speak about these kind of
scaffolded uh developmental capacities
which might involve content as if it
wasn't part of the theory of heart to
mind but it is so I really wanted to
clarify that before this goes uh before

this podcast goes live again uh just more broadly I do want to take responsibility as your podcast host for saying things are accurate and truthful and owning up I make mistakes this was one of those uh places I would have edited out but it really would have messed up the flow of the conversation so I thought it was easier for me to just come on here and clarify that I made a mistake and hopefully uh yeah clear up any confusion again um I don't mind too much if I mess up describing what I think because they're my thoughts but I know that authors far more senior than me have put a lot of time and a lot of effort into their accounts and so it'll be doing a disservice to them disservice to you if I misconstrue or misspeak when describing uh their for and their writings so when I do this again because you know In the Heat of the Moment In the thir of the moment I will make mistakes as I did here and again it's just because these are live conversation so I'm having to think on my feet uh but when I do it again because unfortunately I am fallible please do tell me uh I will always look out for it I'm happy that I spotted this one but again I feel a very strong responsibility to be as accurate as possible so that relies on you as well so please don't inform me for when I make a mistake and I will not keep you any longer because this is a cracking conversation with Chris and I hope you enjoy the rest of

it yeah I'm not very keen on these
ramped up versions um I would it's
conceivable that this might be useful
way of looking at the sort of automatic
mentalizing but I think the sort of
cultural um folk psychology
story very much depends on
representations I mean we almost we're
talking about them as it I mean yeah
um I can't there's a philosopher called
Davidson I can't remember precisely what
he said but it's all about applying you
know rational arguments to what we
believe are these mental States so
you're saying if he believes that the
chocolate is in the cupboard he will go
and look in the cupboard is the sort of
um philosophical approach to this yeah I
mean and that I think yeah no as you say
I mean these are uh sort of an activist
cognition on steroids um there of course
people and I don't think you would
disagree with them that recognize that
if there are representational schema
those are built off scaffolds that occur
from social cognition developmental
psychology being with others and I don't
think you would disagree with that and
that is something I want to touch upon
um in terms of self-modeling and self-
perception but yeah um actually I'll
backtrack I don't think they would say
that uh cognition is non-intentional
just that it's non contentful or non
representation um but there are
obviously quite severe critiques of that
and there probably are representation as
well as they make clear in their works

so now expanding that out to away from
dactic or not dactic diadic interactions
yeah into broader social
contexts why going Beyond just the
coupled Duo why is theory of mind or
mentalizing so important just for social
institutions more
broadly well I I I I would claim that we
don't think that necessarily okay so we
think
that that some people seem to equate
social cognition with theory of mind
right and we do not sure okay let's
clarify that yeah so we think there are
all sorts of social cognitive processes
that are not to do with representing
mental States or whatever and many of
those might be more relevant to larger
group
interactions and we talk about I mean
what's and the original paper that we
wrote in 1999 I think which is sort of
starting of social cognition we had
mentalizing on one side and we had mirror
neurons on the other side because they
had only recently been
confirmed and um mirans are all about
how we can imitate others and how we
get um what's the word contagion from
others so for example emotions are
contagious we tend to feel the emotions
that other people are expressing we tend
to do what other people do um we imitate
their
actions and one very important aspect of
human sociality is overimitation do you
know about yes I I I I've seen some of
it in sort of uh linguistic uh

Innovation where a language learner learning another's language will overimitate and so they'll be sort of uh uh they'll all speak in RP for example even though no one's spoken in RP since 1910 yes but I'm as I've not thought about it so much in that respect but it's more to do with simple actions um like shaking hands is and you have to do it in the right way and the early experiments where they had these um puzzle boxes which you have to solve by going through series of Maneuvers and the trainer puts in a couple of Maneuvers are not actually necessary to open the box but the children all imitate these as part of the that's the overimitation bit so that they do them even though they're not I see I see and the interesting thing is that chimpanzees don't very interesting the idea is this is we're imitating not to get to the reward we're imitating to show that we are doing it properly we are part of the group we are the people who behave like this and they're yes yes so it's kind of in a sense directed The what in active inference terms kind of a higher a higher order prior yes no that's that's very interesting yeah I mean um I guess to return to my initial question yes I I recognize it might be an overextension of theory mind I didn't mean it in terms of everything is Fe of mind that's clearly not the case because then it

would be sort of serious cognitive overload if I was just mentalizing everyone and everything and it sounds borderline schizophrenic to be thinking that you know my chair has intentions but it seems to me that even in making or or or acting to confirm that higher order prior that I am like you and so you're going to like me and and and we're going to be able to establish this recursive relationship I need to have at the very fundament of that a notion that you are an intentional conscious agent which which kind of seems to me to be a very lowlevel form of theory of mind it might be implicit in our interactions but that still seems to be present yes that's a very interesting idea um one of the things that we've written about is sort of addressing this question how do we know when we're want wanting to in competitive situations and which usually means you're playing an economic game like hide and seek yeah um you need to know what sort of agent you're competing against sure and um if it's a very you know it could be a completely random agent that just is biased so it's very easy to beat one of those or it could be a simple learning agent that depends on what's you know the pattern reward in the past and it's fairly easy to beat one of those and the most difficult ones to beat are the ones that are intentional

and are trying to predict what you're going to do as well as at the same time as you're trying to predict what they're going to do like rock paper scissors or or something like that yes and then that's where you get into this in infinite recursion where you say does he have a representation of what I think he's going to do next yeah yeah yeah yeah
yeah and um
what is interesting at least the little bits of information I have it's very difficult to work out that you're competing against an intentional agent from their behavior and therefore we tend to depend on our prior if we think it's intentional agent will behave in such and such a way if we think it's not we won't and there's this nice experience John donzo's group in Paris where he has people he has artificial AG that can do recursive Tom yeah and he has people playing against them you know at various levels you can have zero Tom one Tom two Tom and so on in terms of recursion and if you tell them they're playing against a one armed Bandit they assume that it's not intentional and they lose when the agent is functioning at two two levels of recursion even though they have several hundred
they think it's yes if they think it's a person they will adopt the high level strategy and of course this is

reasonable because this high level of mentalizing is extremely cognitively um demanding yes so you're going to avoid going to that level if can yes and there's this nice and do you know about the beauty contest game which is a nice example of this

I

don't ah this is invented by Jean mayard KES anyway the task is you're part of a group everybody has to produce a number between one and

100 and the person who produces half the mean production will

win

okay and it ke Kane's call it a beauty contest game because when you have to say who is the most beautiful person

it's not who is the most beautiful

person but who people think is the most beautiful person anyway the argument is

you know if if you think about what are

the other you have to think about what

are the other people going to do and you

think if they're completely stupid then

this will be a 50 will be the average

number and therefore you'll win with

25 if they're not completely stupid they

think it's going to be 25 therefore

you'll win with 12 and a half if you've

read about um Nash equilibrium you think

everybody is super rational and you're

say one so you always lose um and the

average I mean there have been huge

experiments on this in newspapers where

they get thousands of people to take

part and the average number level of

recursion is two and a

half interesting yes it has to be a cut
off there has to be a cut off this is
what they call I think car would call it
bounded rationality or something yes
yeah yeah well yes I guess it's also I
guess in some sense to do with the depth
of your temporal hierarchy yeah um
because each recursive step seems to me
at least to be kind of a an N plus one
in your kind of information space and so
you're just going to the next one and it
has to be an end at some point we can't
plan forever that's fascinating okay
that's fascinating yeah no that's that's
the other game I'm interested in is the
um stank Hunt game you know that yeah I
know the fact yeah yeah Y and that's
again seems to be recursive because if I
go to hunt STS I have to believe that
you're going to hunt STS and I have to
believe that you think that I'm going to
hunt
STS and it seems to me theistic there
that soles all this to say we are
Cooperative
people and I don't have to worry about
this yeah yeah a couple things come to
mind there I mean dennet as I'm sure you
know has this notion of an intentional
stance um you think that maps
ontologically onto our sort of priors in
some sense that we we start off with the
notion that things are intentional
animate conscious however you want to
term it and then we start pruning or is
it the other way I think I think it's
the other way around we want to start
with the idea that it's all very easy

yes yeah yes but but we do have I I think we have a prayer that people are intentional and and would that be and would we get that evidence from the phenotype of people so to speak so so if you have the sim a similar body to me you move in a similar way to me you speak in a similar way to me that is evidence to confirm that prior but I guess it has to be an evolutionary prior quite a non conceptual level because Evolution doesn't work like that it's not going to say if you've got if you're six foot you're an agent right it's it needs to be a little bit more subtle than that yeah but certainly there's this um somewhat discouraging work on ingroups and out groups groups are considered to be less intentional gosh yes that's quite terrifying yeah yeah yeah but that said I mean we do have it makes me think of the Fus of form facial area and the fact that that certainly constitutes some kind of perceptual prior at least yeah excellent that's very good um I mean I was listening to an interview with that you and UA did with uh I think it was the denter it would have been about a month ago and I think I I just was listening to the beginning and UTA I think it was UT that we could consider cognition all like entirely social cognition so all cognition is social cognition it makes me think my Master's feis was uh exploring what's called social Baseline Theory and I wonder what you thought

about it I'm not sure you're aware of
social Baseline Theory I'm not so this
is um a kind of the experimental work
for it started in of 2006 2007 and then
the theory was fleshed out sort of early
last decades this people like James
Cohen I think it's Cohen it might be
Cohen um Elizabeth Gross these are
people that the audience can search up
but the notion is is that the primary
ecological niche we find ourselves in is
social yes and so what that means is
that the human brain for adaptive
reasons expects social resources because
it and in doing so it that acts to
mitigate risk and also preserve energy
expenditure yes so what they then say in
white it's calledo social Baseline
theory is that being with others
constitutes a cognitive Baseline so that
when we actually put people in an fmri
scanner we get them to do a task alone
which is the standard in cognitive
psychology we're actually viewing them
as a deficit because they have to engage
in all of this self-regulation
self-regulatory processes that they
don't they would be doing to a lesser
degree if they were have trusted others
how does that sound to you as a a theory
oh I I I I love that I'm very sad that I
didn't know about it it's really it's
it's there's this really the really
beautiful experimental studies I think
the most famous one is that people will
perceive a hill to be less steep if
they're with someone else oh yeah
because they I think the perception in

some ways again this feeds into a broader question about to what degree is perception influenced by top- down factors is it modular I mean I come from active influence I'm saying no like it's definitely influenced but I think notion is a perceptual uh yeah a percept is influenced by your notion of how much effort it is going to be in some sense the affordance how much effort is going to be to exploit that affordance so being with others reduces that effort no I think that I would entirely agree with that and there's other experiments that we did talk about in the book which I didn't know how they relate to this and I'm not sure whether the Baseline people know about this about affordance this is this business if you if you have a table with mugs on it you will you can show that you will get when the mug appears as it were if it's within your reach you will get um motor activity in motor cortex and there's a social version of this if there's a second person sitting at the table even a a mug that's out of your reach but in his reach yeah Will activate the motor system so that you're treating this is sort of group affordance it's I call it the wi mode it's within our reach yeah and that seems to happen automatically so that's entirely consistent with this that is that is wonderful I guess in some sense from an active inference perspective it's worth predicting others actions as

if they were our own because their actions are so influential on on personal outcomes but I was thinking more it's like the monkey and the rake we can use this person as a tool to get yes yeah ah yes ah okay so mine was a bit more generous I guess it was a little bit more altruistic ah okay okay cool yeah that's that that's that's an interesting um that's an interesting stance on it I

guess I mean the answer yeah a test of that would be if the P other person sitting at the table is your enemy would you then not get the affordance from the things yeah has someone done that experiment I don't think so no worthwhile um an interesting experiment thought experiment that I think you've used and Carl has used is let's say we're on some exoplanet and it's just me yeah

yeah um I don't I I want to leave the questions about agency and sense of self for a bit because I do want to get there that's kind of more my speciality the work that I focus on is more to do of self modeling but we say okay all cognition is social cognition it's built into us it's it's our Baseline we have social

priors how what therefore what happens when we start denying evidence for those priors if I'm in the planet which was just n of one and I guess are there differences there for if that was always the case versus I've just been plumped there and I've and now I have a memory

in some sense of being a social creature
which is now being denied to me I think
yes there would be a huge difference um
I would say if you just dropped there I
mean it's a bit like being put into um
solitary confinement I'm sure you will
start thinking about the past or making
up stories or about social interactions
and so on and I was thinking also your
nice point about being an MRI scanner
all by yourself I suspect what most
people do when they're an MRI scanner
all by myself thinking about what on
Earth is the experiment of doing this
for sure I think that when I get x-rays
actually I'm like why are they doing
this with the blocks of lead and stuff
yes if you'd never had social
interactions I would I would think you
would be
well in the extreme I had a strange
interaction long ago with Kristoff cooch
talking about I was trying to talk about
you know what's the function of
Consciousness and I was putting the
argument it's all
social so my argument would be if you
were never had interaction with other
people you would not be conscious I
think Cole has a similar yeah St yes and
what was Kristoff's response to
that know Consciousness is in these
special neurons yeah yeah I don't think
that one's got very far uh sorry
Kristoff you're far more senior than me
but I don't think that one's got very
far and he lost that bet to David
Charmers I think he had to give him that

case of wine yeah so so I think he lost
um well that's a really interesting
point let's let's go there then because
as I said this is really where I want to
go is this
self-modeling yeah and I'm thinking
again this is something that I've spoken
to KL about and I've heard Carl mention
which is that we need in some sense to
be able to infer the agency of others or
their self model to be able to apply to
ourselves a self model um there's a
wonderful paper I like shouting out
papers here so that people can go read
and also just to give credit where it's
You by yakob Hoy um and I think his last
name is Michael I'm not sure what his
first name is and it's called why does
anybody have a self it's anybody there
two words and they talk about sort of
cultural imitation as really being
foundational to the development of a
sense of agency in the sense I'm going
to shape myself model on what I see as
kind of optimal a IC behavior on the
part of others so I guess the question
here is how necessary is this capacity
for inference of the agentic state of
another or the mental states of another
for our own formation of agency and
selfhood I certainly think that selfhood
depends on interacting with
others though I think you could get
quite a long way without doing
mentalizing it could be based on
Behavior yes okay
extent um but I'm one of the things that
we quite talk about quite a lot in our

book is
reputation which is very
important economically I mean The
Economist I mean not um Adam Smith who
invented econom economics talked about
he didn't call it reputation he called I
can't quite remember the esteem with
which other people hold you
in was was more important than wealth
and what the main driving factor in
human behavior is to get the esteem of
others is to have a good reputation in
other words though sometimes money can
be a sort of n proxy for this sure and
they're all sorts of interesting
experiments on this because if you have
a good reputation then people will trust
you they will choose to work with
you and they will inv invest money in
you and all that sort of thing
and then that you have a very
interesting um arms
race with the Free Riders because you
want to have a good reputation but you
want to cheat people at the same time
and then you have to that's why being
being anonymous is jly useful from this
sort of competitive situation because no
one you don't have a reputation that
carries with you and um and it seems to
me the reputation is an important aspect
of the self and of course the reputation
is in is not in you it's in the minds of
others yes exactly but in some sense it
then gets transferred onto you I mean I
think it's quite classic literary Trope
in some sense I've mentioned Luigi
pandelo before I don't know why but he's

come up again here he's got a well very peculiar book called unoo chento one no one 100,000 notion is is that these are different ways of modeling oneself self and what what he goes around doing in this book is protagonist is he he goes around trying to spot himself in mirrors so that he can perceive himself um not so he's trying to catch himself looking at himself not as others would perceive him but sort of in this raw moment of self perception and he eventually dissolves or goes a bit mental yeah it's that's one thing that came up into mind there's another far more folk or or popular culture thing there's a very interesting show on BB on the BBC right now um called the traitors uh which I highly recommend which is it's about sort of 20 people in a castle and three of them are traitors and 17 of them are so-called faithfuls and they have to kind of Guess Who's Who and if the traitors stay in they win all the money if the faithfuls stay in they win all there's some trash reality element to it but it's a very interesting psychological recursive thing where there's lots of triple quadruple bluffing going on so that's really that in in action and reputation yes plays a very big role in trustworthiness and you're watching that as the objective sort of Godlike figure and saying well how could you not see that this person is so malign um but you're not there in some sense okay that's that's interesting I guess a

further question I had and this is actually something I would love to explore personally um because I don't think anyone's done that much work on this um so I put this down as a sort of thing that I might want to explore in the PHD which is that we seemingly have homogeneity in self- modeling um so there are certain things that just come out of the literature who I'm thinking about people like Dan Dan Harvey and Shan Gallagher and um yob that it's in some way narrational in some ways it's uh epistemic it's somewhat coherent um again this goes all the way back to like people like Derek parfit um but it's and and that is homogeneous across us I was do you think that that is quite a thought I had is that that's rooted in the fact that we all engage in similar processes to arrive at that self-modeling to arrive at that self-conception namely inference about others does that seem reasonable yes I'm not sure I mean I would have thought that there's a I mean I'm not sure are you saying that because of our in inherent individual cognitive processes being similar to others we arrive at the same end result in some sense but it's a little bit more specific in the sense that uh let's say I'm a I'm a child and uh I watch my parents enact presumably agentic Behavior yeah my model of agentic behavior therefore

starts to align with theirs yeah and
what I do is I in according to act of
inference I act in order to provide
evidence for that prior
and so you see this kind of um
convergence over time on a self model of
course there's going to be variety but
these core features are
present I would think that certainly has
a role but I think there's also a role
for sort of top down cultural effects so
you're told this is what agents how
agents behave right as I was saying
you're told I mean even the very simple
level you're told that people be you
know when
people are if people are eating a lot it
means they're hungry I mean even at this
simple level our behavior is determined
by our desires is something that to some
extent is top down imposed but I was
thinking you were mentioning all these
various people there's an interesting
narrative story yes he does in and I'm
inclined to I I think there are
different individual differences in this
I have no idea I don't have a plan I
never had a plan for my life as it were
me neither yeah and I I might be a quite
different person next week or and that
was yeah so so yeah Gay's critique is
very apt here um and here I guess he's
he's kind of actually criticizing dennet
because dennet is quite keen on talking
about the linguistic form of this
narrational sense of self yakob how's
got a similar point there which is that

maybe we can replace the word
narrational with it's it's not as catchy
but something like a predictive
coherence or
hierarchical constraints or something
like that in the sense that the higher
levels of your uh generative hierarchy
are tracking lower levels and so your
preferences right like I like ice cream
is going to dictate my behavior in the
moment that I'm more likely to go get
ice cream and that feeds back and
forward right I think that maybe we
could just I don't know what that term
would be um but no I think it's a
reasonable critique cool that that's
interesting yeah let's um let's go to
Consciousness then because it was
something that you brought up um so
there were two quotes that I'll read
back to you if I may that you've said
and I'm curious
about what you think these are from
papers I haven't just I was a completely
different person then yeah yes yes yes
that's very yeah very Derek parfit there
are lots of you um in 1995 this is a
while ago in a very I like the title of
this paper Consciousness is for other
people you wrote the primary function of
Consciousness is to permit high level
interactions with other conscious beings
and then this is so that's about
function yeah and then in another paper
you wrote phenomenological Consciousness
is necessary for taking an intentional
stance towards other agents to do this
we have to be able to treat ourselves as

agents also I think we can just brack
off that second sentence yeah yeah
because we kind of touched upon that but
there's there's one point where it's
function
is interactions and the second one is
almost nature the necessity for uh
intent the the necessity for
Consciousness to take an intentional
stance I don't know which order you want
to take that in but it would be really
interesting to unpick both of
those well I
think we've taken that a bit further I
start with the first one that sure is so
this is this study on
metacognition with various people
particularly Nick Shay and Cecilia Hayes
MH and the idea is and it but it comes
from originally from our the work with
bador which is about people working
together you know this study I'm sure so
people are working together to detect a
weak visual
signal and we show that two people
working together and coming up with a
consensus answer are more accurate than
the more accurate person working on
their own in the right context at least
and the reason that this
happens is that they're able to tell
each other how confident they are in
their in their
belief and on a tri B all basis you
choose the answer of the more confident
person so in other words it depends on
them telling each other about their
phenomenal state if you

like why

yeah I've come across as David charma's
parrot on many times in this in this
podcast and I'm going to do it again why
why is that necessarily the case so why
could it not ju why could it not just be
an offline computation whereby we have
two probability distributions that are
just done completely like offline
non-phenomenal they're just numbers um
and you know you have a higher order
agent or or one of them you know there's
just a there's just a algorithm if
choose the higher one and none of this
is done online why do the lights need to
be on um for that um for that successful
Interac the information has to get from
one mind to the
other

yes yes if if we presuppose that
uh

well what for adaptive
behavior in this example yeah yes
could that not be achieved also with
offline Minds I guess is my point could
that why do we need to invoke like
intentionality qualia subjectivity all
the fundamental tenants of Consciousness
yes I do I don't really quite understand
so CH so Charmers in 1995 speaks about
how you can give me a function of
Consciousness so whether it's uh like a
global workspace or whether it's binding
or whether it's this
and but you can never tell me about how
that function maps onto its phenomenal
nature and why that function couldn't
happen I think what's wrong with that is

he's talking about the individual brain
and I'm talking about interactions
between two people and the key thing of
the interaction is that they have to
communicate with each other in this case
confidence yes but could we not have
something like an artificial agent which
has Precision which could be a proxy for
confidence and they're not conscious
they just have done some mathematical
computations the inverse entropy and
this is my confidence right based on
like you could have that with a large
language model for
example so you would have to have so
your argument be if you had two
artificial agents they could also get
better than one working on their own
precisely in this
situation yeah there you go that that
would be my contention not that I again
that doesn't
discount the that doesn't discount the
putative role of Consciousness it just
discounts or calls into question the
necessity for Consciousness I guess
right but I mean on
the on the basis of that sort of
experiment we went a bit further and
started talking about do we actually
learn from each other how to interpret
the is in the states that we
have um and again it becomes very in
intentional and the in and I'm not sure
if this is relevant or not but when we
talk to each other about how confident
we are we can actually then you get in
competitive situations we

can express more confidence than is actually justified in order to in this experiment was to become an advisor so people would want to take your advice because you seem to be more confident about it there that seems yeah yes my only concern I guess is that um there are these interesting Works coming out about sort of extended Trust In Mind Robot interactions or mind artificial agent interactions and if I have a belief that even a robot is extremely predictable and reliable and productive I can offload um a lot of my capacities to that robot and actually if we were in this kind of task I could very happily take its uh its interpretation of the events or its position as my own and I might receive great outcomes but I don't need to invoke in that any representational space within that robot does that in some ways not call into question even something like confidence metacognition whatever it is again I'm not I'm not calling into question the fact that it might be this it's more the necessity of it right um anyway it's an interesting it's an interesting little rabbit hole

Consciousness does this we've g into you know everyone I've had on and we talk about Consciousness it it leads into these rabbit holes um we had an interesting discussion on Tuesday The Institute of philosophy because they're all obsessed with large language models yes and they

were talking

about does it mean when you know does it
understand what it's

saying and it was interesting that they
the presenter was finishing up by saying
is there is meaning but it doesn't mean
what it says so basically they decided
that to go the the prop you had to have
intentionality

right which large language models don't
currently have how much of this is sort
of rooted in um John SS work Chinese
room and sort of thing yeah excellent
yeah yeah yeah it's good I like to see
that 80s philosophy still has its role
um is what I was sort of brought up on
um excellent let's now it's not a jump
because I I think the wonderful thing
about your work is that although it
seems diverse they all sort of trickle
down in into these fundamental questions
of agency and selfhood um I want to talk
about

schizophrenia yes um and maybe we can
start with the a very interesting
symptom which is delusions of control
yes so you spoke uh you wrote very
extensively on delusions of control so
again people who are not schizophrenic
or don't have a psychological
psychiatric background might not know
what I mean by delusional control so it
might be worthwhile starting there just
outlining what that phrase
means so this is where the patient says
it is not

me doing causing these actions there's
some external agent making me do these

things and they could be quite trivial
things like combing my hair or they
could be
more um dangerous things like attacking
people or something but the idea
is and this is interesting because
there's a
delusion typically is considered to be a
belief so you can say that they
believe that external
agents are causing their
actions but it seems more likely and
early psychiatrists had more or less
stated this is it's not so much that I
believe that external agents are causing
my actions it's I feel as if external
agents are causing my
actions it's not just a belief in the
sort of philosophical sense is something
stronger than that well that's
interesting so it goes beyond the
propositional in some sense and there is
part of a whole flu of symptoms called
passivity
phenomena whereby you can say I'm it's
not my
emotion it's being implanted in me and
the most perplexing one of all they're
not my thoughts
yes I wanted I wanted to get to that
that that is which is De weird yeah very
weird very it might be worth um
distinguishing the uh delusions of
control from something like an alien
hand syndrome just so people don't
that's right might be aware of this
notion of a Wandering hand so the alien
hand syndrome is typically caused by a

brain lesion and that's where the hand starts moving by itself and the patient says I'm not controlling it or I don't have control over it it's doing these things that I don't want it to do but typically the patient doesn't really say there's some alien force right making it do these things and they try to stop it I mean the classic Strange Love in the film if you do I I actually haven't seen that I know I know I'm going to get shout to that in the comments um no pet say is playing amongst many other characters as Doctor Strange Love who's an exi Scientific Advisor in a wheelchair and he keeps making these well it seems to have some strange overlap perhaps with Tourette's um I'm working on again I don't come at me I'm not saying they're the same but it seems like that um I've come up with this term called oppositional self-modeling which is a one level of some prior a one sort of lower level of generative hierarchy you kind of you have a h habitual um action policy so you'll just do it right so there's got to be some part of your motor Loop where you will move your hand or you will do a tick but then you have a higher conceptual level where you really don't want to do it because you aren't the type of agent that does tick or does move your hand randomly so that that's what I mean by sort of the overlap there with Tourette's there's also utilization Behavior after frontal

lesions where people make these habitual responses situations where they shouldn't they can no longer inhibit them okay interesting yes so in terms of delusions of control perhaps you could explain to people where the model that you proposed um for where this might come from at least some of the versions of it in terms of the inverse modeling and forward modeling yeah well I guess it goes back to helmoltz who noted that when we move our eyes obviously the image on the retina jumps about but we don't perceive it jumping about and the idea is that we use what So-Cal coroller discharge so that you can when you send the message to move your eye you also send a message to the visual system so saying there's going to be movement now but it's not caused by something happening in the outside world it's due to my own eye movement and that applies to all sorts of movements and more recently particular say with Daniel walit you have this idea of inverse and forward models so it says when I move my arm I actually have a forward model that tells me what what where my arm is going to be when the movement is finished and what sort of Sensations are going to be accompanying the movement and because I can predict them I can discount them yes so sort of sensory attenuation yeah and um this is the difference between a passive movement and an an active

movement so in a passive movement when somebody else moves my arm I feel all these Sensations because there's no discounting and that's why can't tickle yourself yes and the idea was that in schizophrenia perhaps something went wrong with

this system of control so it literally was the case that when they moved their arm it felt as if somebody else was moving

it yes

they and we showed or Sarah Jane blore showed that they could tickle themselves

oh really I didn't actually see I didn't

see that I'm aware of her work I know

you got you fre all did wonderful work

in this in this domain maybe it's worth

um unpicking exactly what that looks

like technically it's a shame that I

can't get up all of the nice diagrams

because there are some really nice

diagrams but it might be worth um

talking about so people might think um

okay I want to move my hand to grip this

mug yeah this I love Dante mug oddly

enough um because I'm a bit of a Dante

acolyte that's beside the point um and I

can just I I just know what motor action

to employ and I just do it yes I think

you term that I think that might be ter

the inverse

problem what why why is that why is that

why is that sort folk psychology

notion not possible for you know

cognitive creatures like us

as far as I understand it the inverse

problem is that whenever

you you have a goal of Your Action yeah
which is to reach that mug in the
appropriate way but there are an
infinite number of different ways in
which the movement could proceed and the
question is how do you decide which one
to use yes and there are various
arguments like do you minimize the
energy requirement mhm or the distance
or or I
um Daniel walut and I don't know whether
he still thinks this thinks what you
minimizes the endpoint
error there're all so that's the sort of
argument about and also there always
different muscles you can use and it's
all very complicated and luckily we
don't have to think about it we just do
it you but underlying there is something
underlying all that and certain kinds of
lesions you can becomes extremely
difficult to grasp things interesting so
so so there are potentially infinite
ways of uh choosing a suite of sensory
uh sorry motor movements to achieve a
goal so it's kind of computationally
intractable to find a unique sequence or
a singular sequence um however we can
predict the outcome of Ro sequences now
that's a different point yes that the
critical idea with the inverse
model is intractable to work out what
you what's the best thing to do but with
the forward model once you have given a
set of instructions You can predict
precisely where you're going to finish
up okay excellent and so let's so that's
so we've got the inverse problem the

forward model now and if we then fold in what's going to be another kind of inverse type which is the efference copy yeah what's the efference what's the role of the efference copy here as far as I understand the efference copy is just a way of saying we can predict what Sensations are going to arise from this movement and that's the the thing that enables you to stop the image apparently moving about because you're you know what the movement is going to be and if the outcome is the real outcome matches the predicted outcome and everything is fine yes but if it in the case of schizophrenia the prediction is wrong so it doesn't match the outcome so we have to explain this um prediction error and one way of explaining the prediction error is saying somebody move my arm I see and this and does this require us in some ways to go to for insertion to say that yeah action some way many people believe that thought is thinking is an activity this goes right back to Lura I think in Russian psychology um but it's philosophically becomes very complicated because you can say um you have to say I intended to have a thought yes and that's definitely the Buddhist might have a problem with that if if you're thinking about a problem you you would have some idea of what sort of outcome you want to have as you work your way through this problem and

so they are there is room there

for um you know

metacognitive approach to thinking so

you you could certainly say this I have

not succeeded in my thinking task yes

you say I think I have a good even when

you begin you can say I have a good

chance of achieving this thinking task

and I guess might relate that to tip of

the tongue phenomena where you say I

can't remember at this moment but I'm

pretty sure if I try a bit harder I

should get

there so these would be examples of

putting thinking into a sort of active

framework yes there are some very nice

metacognitive models actually or or meta

awareness models with an active

inference so um there's one by L SED

Smith 2021 which is very beautiful and

these really rich Bay graphs with these

hierarchical layers um how so I believe

there are I believe there's good

evidence that people with schizophrenia

do not have major problems in motor

control how is this accounted for oh yes

well that's exactly right there's always

a schizophrenic whenever you come up

with a theory of

schizophrenia they should be much worse

than they actually are yeah yes yes but

the um they do have they have eye

movement

abnormalities sort of smooth Pursuit and

things like that which would be

consistent with failing to predict

properly the consequences of eye

movements I there's a there some nice

experiments from

Germany you can do strain how do this work you can do experiments where you have to track a moving

Target and this means that you your impression is that the target stays in the middle of your field of view and the background

moves though it's the target that's moving and the background is stationary and you can so you obviously have to correct for

the S of the moving background as it were and you can measure how much how successfully people make these

Corrections and schizophrenic patients are less good at doing that so there are some subtle F sure that you can see yes interesting does this actually I I've

got one more question before we go to delusions and hallucinations because I sense those are slightly different yeah

um I mean this is in some sense a delusional belief um so there's some interesting stuff there's some very um there's some very adjacent or or

actually overlapping work in active inference uh about agency and it's it's very much a offshoot of the uh

comparator models and what we've been talking about so and and a lot of this

is actually now been focused on

depersonalization yes so and idea I

really like is comes from George Dean

he's got this idea of an alistic control

model and what that means is that if I

can predict the sensory outcomes of my

actions very similar to what you were

saying and I get a sort of non prediction error right I get the results that I expect I can start to develop the inference that there was there is an agent in my generative model that does these things right and what happens is so let's say let's say now that that's an inference and I need to keep providing evidence for that inference to keep it high Precision let's say now let's say I'm in a a torture chamber this is an example uh torture is a very reliable uh producer of symptoms of depersonalization right say I'm now getting tortured I have a prior that I am the type of agent that should be able to uh do things in to confirm evidence for my own uh model of the world but in the torture situation I might have a desired action but I'm not able to actually conf do it and therefore provide sensory evidence for it right and so what you start seeing here is the disentangling of that very inference that there is some some agent uh at the heart of my generative model does that how does that sound to you does it sound very much aligned with the way you would think about agency yes I no I think that's very interesting um I mean we there certainly I've been very interested in the feeling of being in control and there experiments on that um and I guess the torture situation is where you're totally not in control which we likely didn't normally

experience yes
um I'm just trying to
think yes a lot this goes I mean it's a
very basing story
that the feeling in control of being in
control depends not only on the
intention to do something but also on
the
outcome yes exactly exactly yeah yeah I
I've been writing with car and others on
so-called flow States yeah uh flow sta
surfing rock climbing playing musical
instrument and there although you don't
have this Rich sense of self this
epistemic agent model this propositional
agent you do feel in control yes exactly
and that is exactly as you say because
you are you have this like alignment
between your the predicted consequences
of your action and the actual sensory
outcomes in this very tight motor
Loops um and I think it it's worth
folding in this notion of inter
receptive inference as well so this is
people like Annel Seth and Lisa Felman
Barrett this notion that at the very
core what um Thomas Metson go would call
your minimal phenomenal
self they've argued that this is rooted
in inter successful interceptive
inference yeah either through autonomic
reflex or eating right some actual
volitional action and I guess something
else there is that torture would also uh
deny the possibility for successful
inter receptive inference if I'm not
getting fed I am
weakening I'm I I am

unable that that prior that I need to
have a certain level of glucose is not
going
anywhere um so I think what happens
there is that the the the the agentic
model itself dissolves rather than the
homeostatic requirements yeah so there's
there's there's some really beautiful
work being done here um
excellent before we go on to delusions
and hallucinations uh um I wanted to ask
I so my my S for is into Psychiatry are
limited I I'm more Adept at cognitive
science but I did read over the summer
uh Rd Langs The Divided
self I was curious um he he he's famous
for sort of instituting a more I would
say
empathetic uh and understanding approach
to Psychiatry that these people are not
just crazy right but they are what we
would now say are actually Bas optimal
just confirming priors that don't really
correspond to what we would see as
adaptive Behavior I was just curious
about what your opinions are of whether
you have any of R Lang and what he puts
forward in the div divided
self
I I no I don't have a very high opinion
of R Lang but then this is mostly due to
interacting with other
psychiatrists um and
I I don't think I might have read The
Divided s but I don't think I
did the idea I mean putting it into that
basian story is quite
because that seems a very good idea that

is to say they have the wrong prize but they're behaving I mean certainly all the stuff on delusions yes is the suggests that PE the original idea was that they're irrational yeah whatever that means whatever that means and that in other words they are not argu you know they're not appropriately arguing from the evidence and um all the experiments that ever been done sh they're extremely rational and they're probably more irrational than other people and as you say it's just that they start with the wrong

priors yes there's something in basian modeling called the complete class theorem which is that any any action perception can be cast as Bas optimal just given a certain set of Prior they're slightly worrying yeah slightly worrying well but it accounts for things um it accounts for all I mean basically all phenomena because if you speak to co well actually you wouldn't go that far but you could cast it all as this kind of variational basian inference so it's all Bas optimal everything is BAS optimal um but I guess the the flip side of that the positive side of that is that maybe it does license a more humanistic approach to psychiatric patients in so far as there's this interesting argument I don't know if it's an argument there's an interesting point which is that maybe something like this going slightly away from schizophrenia but it's relevant

something like PTSD or uh OCD or general anxiety disorder actually is could be quite adaptive in a certain ecological niche absolutely if I'm the war zone being you know having hyper salience um and being really you know having very high Precision over the likelihood of some sensory input and downweighting priors like in you know arguably in PTSD is actually really useful um but it's just not adaptive in modern you know calm London on leafy regions Park so how do do you think in your experience in Psychiatry this work has altered the way that we view psychiatric patients yes I think it has um I mean in psychiatry in reality of course it's very difficult because there aren't enough of them and there's no money and they can't find any drugs that work but I mean that's the how that really operates but going back to slightly earlier question people like gosh got his name there was the idea that the symptoms of schizophrenia not understandable which is in another you know they have nothing to do with reality or our lives and I guess one of my AIMS in the sort of story story of delusionist control is to say they are understandable it literally does feel as if somebody was moving my arm and and how much would you say this has come from your it would you say that person yeah yeah do do you sense that this um

empathy that you have in in the very literal sense of empathy that you can that it's understandable comes from direct work you know directly working with these

people oh to some extent yes I mean in the when I was with the medical research Council my lab so all MH was in the middle of the acute psychiatric ward so I was interacting with them every

day and it was quite good because they were actually very bored so they would come and knock on the door and say have you got another experiment for me to do yes I think I think uh that's where K started was in psychiatric rotations yeah yeah how how did you guys first meet if I may ask was it at UCL no we met at the um the psychotron unit at the hospital oh

W where we had a there was a

project I think run by Peter little to basically start scanning patients with schizophrenia to test various theories and I if I remember rightly Carl was appointed because he had the psychiatric background to work on this project and that was in the very early days before we knew how to

analyze brain Imaging data yes so he was given the task of analyzing the brain Imaging data and the rest is history as they say yeah quite quite what was it like uh what was your your most cited paper on ever yeah is the uh statistical parametric mapping yeah work that you

did again we're going a little bit off
piece but this is the beauty of this
podcast like you know we can go wherever
we want what was it like be again I
don't know how SPM works I don't know
how I'm not a brain scanner so maybe one
day um or voxal base morphometry or any
of these things but I'm just curious
about what it was
like no it was very exciting yeah
exciting and I think what was wonderful
about Carl is that I mean there lots of
other people doing Imaging for the first
time and they were tending to invent new
ways of analyzing the data yes whereas
Carl very sensibly said but there are
all these statistical paradigms already
there yeah these are what we should be
using and
that's yes I've always kind of wondered
how Carl knew about them in the first
first
place um but I I sense that your work
on uh forward modeling um should I I you
know I was reading I had the joy of
going not obviously through the entire
catalog of your works because that would
take me until next year but going
through a couple of them again and some
of them are sort of canonical I've read
them before um and it's what struck me
almost like reading Thomas metzinger
I've said this before about metzinger
how incredibly forward like in
retrospect it just seems like you're
speaking the same language as we're
saying what we're saying now in active
inference Landscapes not quite but

nearly not quite but well there's a lot of overlap and it's incredibly influential and I would say that for you and uh Sarah Jane and someone like Thomas metzinger it's it's really remarkable how um you know how much was how much how how in conversation were you with Carl when he was beginning to develop the free energy no all the time we shared an office oh nice so I have lots of passive smoking experience oh yeah he he he does like a cigarette fair enough I think he's I think he's warranted one and um have you given much thought I guess we are on the active inference podcast so I ought to ask something more specific I know we we spoke before an email and you said you know this is not directly your line of work yeah but you were around for when the 2010 paper was released the 2009 paper was released uh I think a free you know 2005 might be the sort of Beginnings um what were your kind of experiences around then and and how did you contribute well I'm not quite sure yes I Carl has his habit of putting my name on papers from time to time and I guess I contribute to some extent by saying we need more independent references or something like that or or I sure sometimes change the wording a bit and know what I was particularly interested is I was very keen and it really goes right back to Richard Gregory and people like that the the approach to

perception and the idea that we if our
you know if there's a prediction
error we have to update our model world
and I thought that was very exciting and
wrote about that in my previous book and
so I was particularly excited with
active inference because that gave you
the other side of the picture and when
we get a prediction error we don't have to
update our model of the world we can
change the world instead isn't that
amazing I thought that was wonderful I
know it is wonderful what what are your
thoughts so this is kind of what if
anyone's familiar this is what we would
call the low road to active inference
yeah this is predictive coding uh Helm
Holan perception and the like and action
as action as inference I guess you could
say yeah have you given any thought to
the so-called high road and by that I
mean dynamical stochastic systems and no
that's too difficult for me I'm trying
to learn the maths yeah you're not wrong
you're not wrong it is rather tricky
excellent um okay so I said we would
talk about hallucinations and delusions
I think delusions we've somewhat touched
upon which is kind of this uh fixity of
priors um that is not allowing for
updating but
I read you know I read kapur's work on
um sort of the dopamine aberrant salience
model yeah I never thought that any of
this stuff really accounted for
hallucinations where where where do you
the best of your knowledge do you think
hallucinations might come from well I I

with Paul fetch we read a paper which we're basically saying there's no fundamental difference between hallucinations and delusions okay if you take a Bayesian approach yes yes because the hallucinations are it's just about perception rather than beliefs but in both cases it's all about whether you update and where the evidence comes from but the thing you have to remember about schizophrenia is these hallucinations are predominantly auditory and they're predominantly hearing voices and One symptom is so all thought or no yes you can you I hear my thoughts as if they're being spoken aloud um and you can even have a sort of very similar story with the forward model and there's work on this so that when you speak you hear your own voice but you suppress it and there's some evidence that when patients speak that again the suppression system seems to go wrong in relation to speaking as well and you have inner speech and all these sorts of things so the one idea is that they're perceiving their own inner speech as if it was from an external source which is not that far away from a delusion of control yes I sometimes how true is that if when one meditates for example if one is doing vasna um it does seem like those

thoughts are coming from not from
you so this might be looping back into
our sort of humanistic approach to
psychiatrist all so don't tell that to
your psychiatrist yeah yes yeah yeah
yeah yeah I won't I it's
true at least and and the B you know
there's this these very odd overlaps
between Buddhism and Psychiatry and ego
death and all of these kind of things do
you think that this this notion that
selft talk let's say let's stick with
that because it's very interesting is
selft
talk from a Buddhist from a very raw
Buddhist perspective that doesn't get
off the ground because there is no
self these thoughts are just popping
into awareness yeah and it's it's all
process it's process ontology it's not
there's no substance there it's it's
coming and going it's just the field of
awareness so is it just an Adaptive tool
for us to presume or or have this
illusion that it's
self-generated I guess that depends what
you mean by self-generated I mean well
not self-generated but it's what does
selft talk mean then that maybe
that's because it's yes I mean clearly
it's not I'm not intention intentionally
saying these things per say but I'm
perceiving them as belonging to me yes
they're not coming from outside is yeah
sure but what's in this in this way we
have to kind of ask well what is inside
and outside yeah so so well that's I
mean that's a common one I mean one

quite dominant theory of schizophrenia
they've lost the boundary between inside
and outside that's really interesting
they've lost their Markov blanket yes if
you yeah yes but I guess I guess the
reality is you never lose a markup
blanket I mean this to be honest it it
points to quite it points to theoretical
difficulties in Act of inference which
is something that's been brought up in
this podcast before and in in other
works which is where do you even cast
the markof blanket and what is it to say
that you have internal States and
external States really what those
actually mean um and I give credit to
Maxwell RAM for pointing this out to me
is that internal you can just say are
tracking and external are
tracked but that said you have someone
like Andy Clark who might say well in my
tracking mechanism I might have my
telephone or my notepad and then you
have yob how and says no no no the
evidential boundary is the mind and even
the body is being predicted yeah does
any of those uh dichotomies or debates
inform something like um what we've been
talking about selft talk and uh
delusions of
control well selft talk inner speech
people used to call it
yes and that is and again the Russian
psychologists I mean voty I think but
and there's some evidence for this that
you know young children particularly
doing difficult tasks what we might call
frontal tasks you actually talk aloud I

must now do this and I must now do this
and this becomes internalized so you're
do that internally and I think for some
difficult tasks and things like mental
arithmetic there is indeed in the
speech which is being used to
control what you're doing interesting
and um some schizophrenic
hallucinations where the where they hear
a voice commenting on their action so
now he's going to do this and then he
why is he doing that
and it's all very weird what they but
you can see a sort
of
thread coming through this I mean that's
what I would call selft
talk and self is
not important well I wonder whether
there is some uh some space for a
predictive model here yeah in so far as
if we're going to talk about what we
meant by self self-generated an agency
we were talking about uh deploying some
prediction and having in association
with that prediction uh not just I don't
just deploy an action policy I also uh
precise predictions about the sensory
outcomes I'm expecting to receive and
perhaps we can talk about selft talk in
a similar way which is that at the
Inception of my selft talk let's say I'm
predicting uh the Sens or sensory in uh
Speech marks consequences of an action
policy of saying this at $t+1$ yes and
then I receive that back and that seems
self-generated yeah yeah and I guess if
that mechanism itself is disrupted

through synaptic problems Precision

waiting

problems then that is not only just

going to be motor action but also mental

action yeah is that appropriate

absolutely yeah maybe I should be a

psychiatrist mental action is a good

term yeah yeah mental action is a very

good term well I mean all of this um

it's it's kind of become very clear to

me that

all such a plethora of behavior in in

active inference is mental action I mean

the very deployment of precision uh um

over what now you know we're using these

basian graphs these So-Cal pom DP

schemes so you have your sensory

expectations your C Matrix your uh State

transitions your B Matrix these are

deploying precisions over those are all

mental actions yeah um so okay excellent

okay that's that's okay interesting okay

that's very that's very cool um I was

wondering so there's a uh I believe he

was a philosopher called Lou sass or maybe

he was a psychologist and he wrote a

book called madness and modernity he yes

was he more sort of

modernism it might be it might be it's

been a while since I read it yeah um

well please please educate me because

I'm yeah I mean I read it too but it was

a very long time ago um so I mean it's a

removal of sort of structures and Bas um

um standard procedures and

yeah stream stuff um and how does that

feed into schizophrenia schizoph is sort

of uh the iology of schizophrenia

somewhat linked to having an unstable lifestyle um yes I mean there's an interest this is very um unlikely to be true but there's interesting stuff on volatility so with when the when the environment is volatile you should have a faster learning rate yes and I think that somehow interacts with the dopamine system okay so you could and there is some evidence that children brought up in very volatile environments are more at risk and one wonders whether and then you've got this dopamine aspect coming in as well if we've got too much dopamine so you get the there's that sort of story and modernism in the sense is increasing the volatility because you no longer rely on um I see cultural Givens you know currently an interesting idea in relation schizophrenia which is bringing the cultural thing is that most of our beliefs at this high level are constrained by interactions with other people and culture more generally and you could argue that what's odd about what the reason that schizophrenic beliefs don't update is because they're no longer constrained by this high level cultural if what other people say ah so these are yes your work is all coalescing exactly that's cool that's nice yeah that's very yes it's interesting isn't it because we think of ourselves as

these we live in a rather
individualistic Society where we think
of ourselves as these atomist IC beings
and we're not and we're
not that's terrifying
um good excellent so so the book came
out last year yes what's on the what's
on the horizon for Chris FFF boa FFF the
fffs well um we're in our 80s now um you
wouldn't be able to
tell um we haven't got a major
project in line
um we have a quite exciting interaction
with our friends in orus in Denmark yes
where they have uploaded all our papers
and are trying to construct a large
language
model interesting who we really are
interesting uh is this Andreas
well no it's in his his yeah yeah yeah
yeah because you have a you have a
partnership with him or is that still
Ono we visit every year yes yes
excellent he's great I think he might be
coming on um yes you should certainly
talk to him yeah no I'm extremely
interested in this work Chris This was
um this was a delight um it's been a
delight on my part to go back and
read uh your texts I mean I did finished
my Master's just background about myself
in case you didn't I finished my masters
at UCL in experimental psychology last
year and I would say sort of your
presence and K's presence sort of
lingers in a positive way not in a sort
of ghoulish way through the corridors um
so thank you y my pleasure it's been an

absolute pleasure all right excellent